

Aakash Kumar Agarwal

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PROFILE

AI/ML Engineer and MS by Research graduate from IIT Bombay with experience building **LLM, multimodal, and machine learning systems**. Hands-on with model fine-tuning, post-training, evaluation, agentic workflows, data pipelines, and ML experimentation using **PyTorch, Hugging Face, LangGraph, FastAPI, Docker, and SQL**. First/equal-contribution author of papers at **AAAI 2026** and **EMNLP Main 2025**.

EDUCATION

Indian Institute of Technology Bombay

MS by Research, Computer Science (AI/ML); CPI: 8.83

Mumbai, India

2023 – 2026

- **GATE Computer Science 2023**: All India Rank **156** among 75,680 candidates.

TECHNICAL SKILLS

Programming & Backend: Python, C++, SQL, FastAPI, REST APIs, Git/GitHub, Linux/Bash

ML & LLM Engineering: PyTorch, Hugging Face, Transformers, Scikit-learn, NumPy, Pandas, Fine-tuning, LoRA/PEFT, DPO/GRPO, RAG, LLM Evaluation, Multimodal LLMs

AI Systems: LangChain, LangGraph, AI Agents, MCP, Tool Calling, Model APIs, Embedding/Retrieval Pipelines

Infrastructure: Docker, Model Training & Inference, Experimentation Pipelines, Data Processing Pipelines

EXPERIENCE

Fujitsu Research

AI/ML Engineer Intern

Bangalore, India

May 2025 – Jul 2025

- Designed and implemented **Hypothesis-Driven Reasoning (HDR)**, an iterative generate-and-verify system for improving LLM reasoning over multimodal observations.
- Built multi-stage LLM workflows using **Python, LangChain, and LangGraph**, integrating inference, verification, memory consolidation, retrieval, and experience transfer.
- Built automated evaluation pipelines across GPT-4o, Qwen2.5, and Gemma3, achieving near-oracle performance and up to **56.7% improvement** over baseline systems.

CFILT Lab, IIT Bombay

AI/ML Engineer

Mumbai, India

Jan 2024 – 2026

- Built LLM and NLP systems for **machine unlearning, multilingual LLM safety, multimodal reasoning, and large-scale social-media analysis** using PyTorch and Hugging Face.
- Developed data processing, model training, and evaluation pipelines involving multiple foundation models, large datasets, and automated benchmarking.

Accenture

Software Developer

Gurugram, India

May 2019 – Jul 2021

- Developed and maintained production software applications across requirements analysis, implementation, testing, debugging, and deployment.

SELECTED PROJECTS

Machine Unlearning for LLMs via Hypernetwork-Generated LoRA

IIT Bombay / IIT Kanpur

- Built **HyperLoRA**, a PyTorch-based system that dynamically generates per-shard LoRA adapters from learned embeddings, enabling modular LLM adaptation and $O(1)$ shard deletion while keeping the base model frozen.
- Implemented and benchmarked five adapter routing/composition strategies with LLaMA-3.2 on TOFU; deployable routing achieved **0.80 AGG** versus **0.81** for retraining from scratch and **0.53** for the strongest tuned post-hoc baseline.

ReDepress: Longitudinal NLP Pipeline for Depression Relapse Detection

IIT Bombay

- Built an end-to-end ML pipeline over **10 years of Reddit data**, including ingestion, filtering, LLM-assisted labeling, feature extraction, temporal preprocessing, model training, and evaluation.
- Trained transformer-based temporal models over longitudinal user histories; cognitive features improved accuracy from **0.81 to 0.88** and F1 from **0.76 to 0.84**.

SELECTED PUBLICATIONS

- **Aakash Kumar Agarwal**, Om Chaudhari, Nihar Ranjan Sahoo, Abhinav Joshi, Ashutosh Modi. **Learning to Forget: Hypernetwork-Driven Sharded Unlearning in LLMs**. *ACL ARR 2026, under review*.
- **Aakash Kumar Agarwal***, Moyuru Yamada*. **Hypothesis-Driven Reasoning for Large Language Models**. *AAAI 2026 Main Conference*. *Equal contribution.
- **Aakash Kumar Agarwal***, Saprativa Bhattacharjee*, Mauli Rastogi, Jemima S. Jacob, Biplab Banerjee, Rashmi Gupta, Pushpak Bhattacharyya. **ReDepress: A Cognitive Framework for Detecting Depression Relapse from Social Media**. *EMNLP Main 2025*. *Equal contribution.